

CHANDRA MOULI DASARI

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AWS Certified (Star Cloud Computing) | Published Researcher - ICET 2024

PROFESSIONAL SUMMARY

M.S. Computer Science candidate at the University of Florida (GPA 3.79/4.0) with 1+ year of industry experience engineering production data systems at HCL Technologies. Designed normalized PostgreSQL schemas across 6 tables and 31,000+ records, authored PL/pgSQL stored procedures, and built Python ETL pipelines that cut manual processing by 40%. Seeking to apply database engineering, automated reporting, and AI integration expertise to FDOT's Full-Scale Pavement Testing Program.

TECHNICAL SKILLS

Database Engineering: PostgreSQL, MongoDB, Schema Design, Normalization, Complex Joins & Subqueries, PL/pgSQL Stored Procedures, Indexing, Views, SQL query optimization
Data Engineering & Automation: Python (Pandas, NumPy, SciPy), ETL pipelines, CI/CD, Git, Linux, AWS (EC2, S3), automated reporting
AI / Machine Learning: PyTorch, TensorFlow, scikit-learn, CNNs, U-Net, Regression, Classification, Anomaly Detection, NLP (TF-IDF, NER)
Visualization & Reporting: Matplotlib, interactive dashboards, data governance documentation, reproducible workflows
Cloud & Certifications: AWS Certified - Star Cloud Computing (2022), HIPAA-compliant data handling, version-controlled deployments

PROFESSIONAL EXPERIENCE

Software Developer | HCL Technologies | India May 2023 - May 2024

- Designed and maintained 6-table normalized PostgreSQL database handling structured operational data; wrote complex multi-table JOIN queries, subqueries, and aggregations - directly replicable for FDOT pavement sensor data registries.
- Engineered automated Python (Pandas, NumPy) ETL pipelines processing 1M+ records per run, reducing manual preprocessing effort by 40% and enforcing full analytical reproducibility across varied dataset schemas.
- Developed ML models for classification and anomaly detection using scikit-learn; improved F1-score by 18% through structured cross-validation and hyperparameter tuning across 5+ model variants, with full precision/recall reporting.
- Deployed scalable microservices on AWS (EC2, S3) with CI/CD pipelines, achieving 99.9% production uptime; enforced version-controlled, reproducible ML deployments across all releases.
- Applied NLP (TF-IDF, NER, tokenization) to extract structured insights from unstructured text corpora, reducing analyst review time by 30% - methodology transferable to automated pavement inspection report parsing.

RESEARCH & TECHNICAL PROJECTS

NBA/ABA Historical Database - PostgreSQL System Design

- Engineered a fully normalized 6-table PostgreSQL schema (teams, players, draft, coaches, career stats, season stats) ingesting 31,000+ records via automated \copy pipelines - architecture directly transferable to FDOT pavement sensor data warehousing.
- Authored 12+ complex analytical SQL queries including multi-table JOINS, correlated subqueries, EXISTS/NOT EXISTS logic, and ALL-clause set comparisons to answer cross-entity research questions across 3,500+ player and 8,500+ draft records.
- Built two PL/pgSQL stored procedures (seasons_played() and point_matrix()) with dynamic record iteration, conditional logic, and SETOF return types - demonstrating production-grade database programmability aligned with FDOT's analytical tooling requirements.
- Applied ANOVA and regression-style aggregations to identify statistically significant performance trends across player cohorts, simulating the outcome-registry analysis used in large-scale infrastructure research programs.

Automated Data Visualization & Reporting Pipeline

- Built interactive reporting dashboards in Python (Matplotlib, Pandas) delivering automated summaries from raw structured datasets - replicating the continuous reporting architecture required by FDOT's pavement testing program.
- Implemented data governance documentation covering 100% of pipeline stages (data dictionaries, transformation logs, workflow diagrams), enabling full audit trails and reproducibility for research stakeholders.

Encrypted File Chunking System [Published - ICET 2024]

- Presented at the 12th International Conference on Contemporary Engineering and Technology (Samarkand State University collaboration); published research on AES-encrypted file-chunking with metadata management, demonstrating secure, governance-compliant data storage practices aligned with federal research standards.

EDUCATION

M.S. Computer Science - University of Florida, Gainesville, FL GPA: 3.79 / 4.0 | Aug 2024 - May 2026
B.E. Computer Science - Sathyabama University, Chennai, India GPA: 8.78 / 10.0 | Aug 2020 - May 2024

CERTIFICATIONS & RECOGNITION

- AWS - Star Cloud Computing Certification** (Cert No. RiiVQCEAAN, issued Mar 2022) - cloud infrastructure, deployment, and resource management
- Published & Presented** at 12th International Conference on Contemporary Engineering and Technology 2024 (ICET 2024), Samarkand State University collaboration